

A Zero-Trust Edge Intelligence Framework for Post-Quantum IoT Failure Prediction and Merkle-Anchored Auditing

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Abstract—Predictive maintenance for IoT fleets is safety- and mission-critical, yet most pipelines forward raw telemetry to a centralized cloud, adding latency, a single point of failure, and exposure to quantum-capable adversaries. We present a zero-trust, edge-native framework unifying failure prediction, post-quantum confidentiality, and tamper-evident auditing: an edge classifier scores device health and triggers hybrid post-quantum encryption – a ring-Learning-With-Errors scheme wrapping an AES-256-GCM key – only for failure-prone devices, while every event’s hash is committed to a Merkle-anchored, hash-chained ledger regardless of encryption status. A logistic-regression predictor reaches a mean AUC of 0.939 at 0.066 ms latency; selective encryption cuts overhead by 87–95% versus encrypting all telemetry; comparison against RSA-2048 and ECDH P-256 quantifies quantum-resistance cost; and Merkle-root anchoring keeps on-chain storage constant at 32 bytes per batch, over 4,300× smaller than raw payloads at 800 devices. Tamper-detection tests confirm forged events, altered blocks, and replays are rejected, and an ablation study isolates each component’s cost and security contribution. A security analysis states the scope: post-commitment tampering is detected, but a compromised edge node fabricating data before commitment is not, and a missed prediction is a confidentiality exposure – the contribution is architectural, not a new cryptographic primitive.

Index Terms—Internet of Things, Edge Computing, Failure Prediction, Post-Quantum Cryptography, Blockchain, Zero-Trust Security

I. INTRODUCTION

IoT fleets are now central to cyber-physical systems across healthcare, manufacturing, energy, and transportation, where undetected device degradation causes direct financial loss or safety risk. Edge- and cloud-based analytics has improved failure-prediction accuracy and timeliness [3, 4], while a separate body of work studies post-quantum cryptography (PQC) for IoT [16], largely in isolation. Cloud-centric analytics adds latency and a central point of failure; edge-centric de-

signs remove that bottleneck but often treat security as an afterthought. Few studies place post-quantum protections inside a failure-prediction pipeline [12, 18]; most IoT security analyses still model only classical adversaries [5]; and centralized trust schemes introduce their own scalability limits [9, 11]. No framework we are aware of jointly supports real-time, edge-resident failure prediction, tamper-evident data handling, and forward-looking cryptographic resilience in one architecture.

Closing that gap raises four tensions: prediction must fit edge compute/memory budgets [5, 10]; telemetry must be protected without encrypting every message, most of which is not security-relevant; audit logs must stay tamper-evident yet cheap enough to sustain continuously, since raw telemetry on a blockchain is neither fast nor economical at IoT data rates [2, 12]; and PQ primitives must be lightweight enough not to erase the edge’s latency benefit. Our contributions:

- A four-phase, edge-resident architecture predicting device failure from telemetry windows, encrypting only failure-prone devices’ telemetry via hybrid PQ primitives, and anchoring tamper-evident proofs of every event to a hash-chained ledger.
- A conditional encryption mechanism, threshold-gated on the failure indicator, cutting cryptographic overhead by up to 95% versus encrypting all telemetry, without audit-continuity gaps.
- A Merkle-root anchoring scheme keeping on-chain storage constant regardless of fleet size, backed by an end-to-end reference implementation used to obtain every number in this paper.
- An evaluation of prediction accuracy, encryption overhead,

scalability, and a measured comparison against classical baselines, stating explicitly what is measured versus modeled.

II. LITERATURE REVIEW

Edge-based failure prediction. Cloud-centric IoT failure prediction achieves strong accuracy, but round trips to a centralized service add latency and limit scalability. Edge inference reduces this [3, 4], though much of this literature treats confidentiality and cryptographic resilience as secondary. Resource-aware ML pipelines are similarly used across other constrained domains – fraud detection [32], plant-disease detection [33], traffic forecasting [34], obscured-object classification [35], and aerial waste detection [30] – underscoring the broader demand for efficient, deployable models.

Post-quantum security for IoT. Work on lattice- and hash-based primitives for resource-constrained IoT spans smart grids [10], agricultural sensing [21], 6G smart cities [15], digital-twin healthcare [13], and medical IoT [11, 22, 16, 8]. These typically evaluate PQC primitives in isolation rather than inside a complete prediction-and-auditing pipeline [19]; surveys note most deployed systems still assume classical adversaries [5, 1].

Blockchain-based trust and auditing. Trust- and behavior-aware mechanisms flag anomalous device communication [11, 14], but many rely on a centralized analysis point. Blockchain-based approaches distribute trust across peers [2], including federated industrial-IoT trust management [20]; raw telemetry on-chain is storage- and throughput-limited at IoT data rates [12], and most proposals treat failure prediction as orthogonal to ledger design. A parallel line of work targets trust at the consensus layer itself – reputation- and cluster-based miner-trust models [23, 24, 25, 26, 27, 28, 29] and RL-based smart-contract vulnerability repair [31] – addressing trust among miners or contract code, not per-event IoT telemetry tamper-evidence, the layer our ledger targets.

Positioning. Our work combines these threads: an edge-resident predictor decides, per device and per cycle, whether post-quantum encryption is warranted, and only a constant-size Merkle root – never raw telemetry – is committed on-chain regardless of encryption status. Section V shows this changes the measured latency, storage, and encryption-cost profile relative to always-encrypt or always-commit-raw-data baselines.

III. METHODOLOGY

A. System and Threat Model

The system has three cooperating layers at the edge: edge intelligence (local failure prediction), a cryptographic layer (selective telemetry protection), and a blockchain-anchored verification layer (tamper-evident auditing without storing raw data on-chain). “Blockchain-anchored” names the architectural pattern (an append-only, hash-linked commitment structure); our implementation is an in-process, single-writer hash-chained ledger, not a deployed consensus network (Section C). IoT devices stream telemetry to a co-located edge node, which runs the predictor, encryption module, and Merkle-tree construction; a monitoring service verifies inclusion via Merkle proofs. Fig. 1

summarizes the architecture.

We adopt a zero-trust posture: no network link, relay, or edge peer is trusted by default. The adversary can eavesdrop, intercept, replay, or modify data in transit and may possess a scalable quantum computer running Shor’s and Grover’s algorithms. Edge-node compromise is scoped precisely in Section C: a compromised node is detected *after the fact* via broken hash-chain linkage, but one compromised *before* computing a batch’s root can produce an internally-consistent, falsified commitment – so “zero-trust” here means minimizing trust in the network and any single message. Within that scope, the framework preserves confidentiality of flagged telemetry, integrity against post-commitment tampering, replay resistance, and rejection of unauthorized ledger writers (Sections F, VI).

B. Preliminaries

Table 1 summarizes the notation used. Confidentiality resists a quantum-capable adversary via a hybrid construction over $R_q = \mathbb{Z}_q[x]/(x^N + 1)$: a public-key step based on ring-Learning-With-Errors (ring-LWE) – the Lyubashevsky-Peikert-Regev cryptosystem [36], generalized by module-LWE schemes such as CRYSTALS-Kyber [7, 16] – establishes a symmetric session key, which then encrypts the payload under AES-256-GCM. With χ the centered binomial noise distribution ($\eta=2$):

$$\text{KeyGen: } a \leftarrow U(R_q), s, e \leftarrow \chi, \quad b = a \cdot s + e$$

$$\text{Enc}(pk=(a, b), m) : r, e_1, e_2 \leftarrow \chi, \quad u = ar + e_1, v = br + e_2 + m \lfloor q/2 \rfloor$$

$$\text{Dec}(sk=s, (u, v)) : \hat{m} = v - su \pmod{q}, \text{ round each coeff. to nearest of } \lfloor q/2 \rfloor$$

where $m \in \{0, 1\}^N$ encodes up to N message bits one-per-coefficient, letting a single (u, v) ciphertext carry the entire 256-bit AES key; all products are computed via NTT in R_q (Section E). AES-256 is treated as already quantum-resistant since Grover’s algorithm only halves its effective margin (256-bit to 128-bit) [7, 17]. Authentication and replay resistance are enforced at the edge’s ingestion boundary, detailed in Phase 3 below.

C. Proposed Framework

The framework is a four-phase edge pipeline – initialization, prediction, secure event handling, and integrity auditing – so prediction and security enforcement are stages of one per-cycle computation, not separate subsystems.

Phase 1 (Initialization). The edge node trains $F : \mathbb{R}^m \rightarrow [0, 1]$ on historical/simulated telemetry and selects the decision threshold

$$\theta = \arg \min_{\theta} (\alpha \cdot FP(\theta) + \beta \cdot FN(\theta)), \quad (1)$$

where $FP(\theta), FN(\theta)$ count false positives/negatives and α, β weight their relative cost (Section IV). The node also generates post-quantum key material $K_{QR} = \{pk_i, sk_i\}$; both steps are infrequent.

Phase 2 (Prediction). For each incoming $D_i(t)$, the node computes a health score $P_i(t) = F(D_i(t))$ and a binary failure

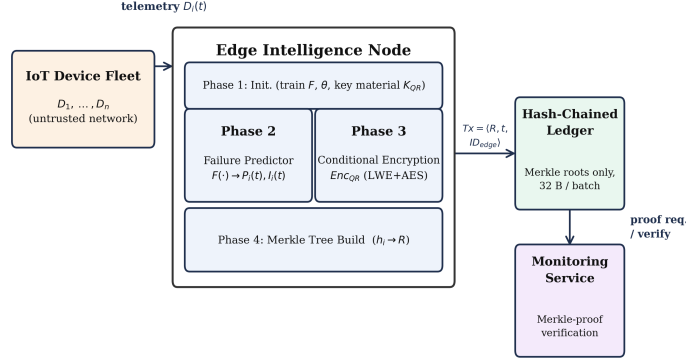


Figure 1: Architecture of the proposed framework: telemetry is processed and, when flagged, encrypted at the edge; only Merkle roots cross the trust boundary onto the ledger.

Table 1: Notation used throughout the paper.

Symbol	Meaning	Symbol	Meaning
$D_i(t), F(\cdot)$	Telemetry vector; failure predictor	$P_i(t), \theta$	Health score $F(D_i(t))$; threshold (Eq. (1))
α, β	FP/FN cost weights	$I_i(t)$	Failure indicator (Eq. (2))
$H(\cdot)$	Hash function (SHA-256)	Enc_{QR}/Dec_{QR}	PQ hybrid encryption/decryption
$R_q = \mathbb{Z}_q[x]/(x^N + 1), N, q$	Ring-LWE ring; dim./modulus (256, 7681)	$\chi, \eta, a, s, e, b, r, e_1, e_2, u, v$	KeyGen/Enc intermediates (Eqs. in Sec. B)
$C_i(t), h_i(t), R$	Ciphertext; event hash; Merkle root	$Tx = \langle R, t, ID_{edge} \rangle$	Ledger transaction
$\nu_i(t), \sigma, (sk, pk)$	Nonce; Lamport signature, keypair	n, m, k	Fleet size, feature count, batch size

indicator

$$I_i(t) = \begin{cases} 1, & \text{if } P_i(t) < \theta \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

available without a round trip to any centralized service (quantified in Section D).

Phase 3 (Secure Event Handling). Admission runs two checks in fixed order: the node verifies a Lamport one-time hash-based signature [6, 8] over the payload bound to its nonce $\nu_i(t)$ and timestamp t , rejecting a forged or mismatched signature immediately (`invalid_signature`); a validly-signed event is then checked against the nonce-cache-plus-freshness-window rule, rejected if t is outside the window or $\nu_i(t)$ was already admitted (Section F, D). Encryption then applies conditionally on $I_i(t)$:

$$C_i(t) = \begin{cases} Enc_{QR}(pk_i, D_i(t)), & \text{if } I_i(t) = 1 \\ \emptyset, & \text{if } I_i(t) = 0, \end{cases} \quad (3)$$

so cryptographic cost scales with devices actually flagged, not fleet size (Section B); the consequence when $I_i(t)$ is wrong is examined in Section A. Every event is hashed, $h_i(t) = H(C_i(t))$ (or $H(D_i(t))$ when $I_i(t) = 0$), and a batch's hashes form a Merkle root $R = \text{MerkleRoot}(h_1, \dots, h_k)$. Algorithm 1 summarizes Phases 2-4.

Phase 4 (Integrity Auditing). Rather than committing per-event data, the node commits only the batch root as $Tx = \langle R, t, ID_{edge} \rangle$ appended to the hash-chained ledger. Verifying inclusion requires only recomputing $H(C_i(t)) = h_i(t)$ and

Algorithm 1 Edge-Resident Prediction and Selective PQ-Encryption Cycle

Require: Trained predictor F , threshold θ , key material $\{pk_i, sk_i\}$, batch $\{D_i(t)\}_{i=1}^k$
Ensure: Ledger block committing the batch's Merkle root

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1: for  $i = 1$  to  $k$  do
2:    $P_i(t) \leftarrow F(D_i(t))$                                       $\triangleright$  Phase 2
3:    $I_i(t) \leftarrow \mathbb{I}[P_i(t) < \theta]$ 
4:   if  $I_i(t) = 1$  then
5:      $C_i(t) \leftarrow Enc_{QR}(pk_i, D_i(t))$                         $\triangleright$  Phase 3
6:      $h_i(t) \leftarrow H(C_i(t))$ 
7:   else
8:      $h_i(t) \leftarrow H(D_i(t))$ 
9:   end if
10: end for
11:  $R \leftarrow \text{MerkleRoot}(h_1(t), \dots, h_k(t))$                   $\triangleright$  Phase 4
12: commit  $Tx = \langle R, t, ID_{edge} \rangle$  to the ledger

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checking its Merkle proof against R ; since only a fixed-size root is committed per batch, ledger throughput and storage no longer scale with telemetry volume (Section D).

D. Correctness and Soundness

Correctness requires $P_i(t)$ to be deterministic given fixed model state, and every honestly-generated event to verify against its committed root, following from F being stateless and the collision resistance of $H(\cdot)$. Soundness requires verification to reject any event altered *after* commitment, following from pre-image/second-pre-image resistance of $H(\cdot)$ and the ledger's hash-chaining (Section C states what this does not cover).

Both are tested empirically in Section F: LWE+AES round-trip correctness, Merkle-proof correctness across tree sizes 1–129, forged-event rejection, altered-block detection, and replay/staleness rejection.

IV. EXPERIMENTAL WORK

We implemented all four phases in Python and used the resulting reference implementation to produce every number in Section V; it will be released as a public repository upon acceptance.

Telemetry generator. We evaluate on a documented synthetic generator rather than a public dataset, so failure labels are known exactly. It simulates $n = 200$ devices, each emitting $m = 4$ features (temperature, voltage, vibration, latency) over 200 timesteps; 30% undergo a gradual sigmoid-shaped drift toward a degraded state at a randomized onset (50–85% of the trace), modeling bearing wear, battery fade, and thermal drift, while the rest stay nominal. Features are rolling-window statistics (mean/std, 5-step window); splits (70/30) are by device, stratified by failure status.

Failure predictor. We evaluate logistic regression and a shallow random forest (50 trees, depth 6) – small models consistent with edge budgets. The threshold uses $\alpha = 1.0, \beta = 2.0$ in Eq. (1), weighting missed failures twice as costly as false alarms; θ is selected on the training split, inside model fitting, and fixed before test-set evaluation – no leakage.

Post-quantum primitives. Enc_{QR} is a from-scratch ring-LWE scheme over $R_q = \mathbb{Z}_q[x]/(x^{256} + 1)$, $q = 7681$ (NTT-friendly, $2N \mid (q-1)$), centered binomial noise ($\eta = 2$), Kyber-style sampling; one ciphertext packs an entire 256-bit AES key, unlike bit-by-bit plain-LWE. The NTT is cross-checked against an $O(n^2)$ schoolbook reference (Section F), *not* a certified library such as liboqs’ Kyber; ($n=256, q=7681$) targets an estimated 64-bit classical/58-bit quantum level [37], below Kyber’s 128-bit target. Scaling to Kyber-strength is a direct dimension increase ($n = 512/1024$), no algorithmic change.

Replay defense and classical baseline. A nonce-cache-plus-freshness-window guard (5 s) rejects any event with a seen nonce or out-of-window timestamp. To quantify the cost of quantum resistance, we implement the same hybrid pattern with RSA-2048-OAEP and ECDH-P256+HKDF from the audited cryptography library, both broken by Shor’s algorithm and included only as the “what we are replacing” baseline.

Ledger and methodology. The ledger is an append-only, SHA-256 hash-chained structure implemented in-process. Prediction results (Section A) are mean $\pm$ std over 10 independent seeds, each re-generating the fleet, re-splitting, and re-fitting from scratch. All quantities are wall-clock except cloud round-trip time (Section D), modeled log-normal near 70 ms, consistent with typical WAN/cloud-API latency.

V. RESULT ANALYSIS

All results are produced by the implementation of Section IV. Every ratio quoted below (e.g., $4,300\times$ on-chain storage savings, $50\text{--}1,000\times$ edge/cloud speedup) is computed from the underlying full-precision measured means, not the rounded table

Table 2: Failure-prediction performance, mean $\pm$ std over 10 independent seeds, 12,000 held-out telemetry windows per seed.

Model	AUC	Prec.	Recall	F1	Lat. (ms)
LogReg	0.939 $\pm$.006	0.953 $\pm$.019	0.775 $\pm$.010	0.855 $\pm$.010	0.066 $\pm$ 0.001
RF (50,6)	0.929 $\pm$.008	0.940 $\pm$.019	0.766 $\pm$.009	0.844 $\pm$.010	2.951 $\pm$ 0.136

Table 3: Selective vs. encrypt-all ring-LWE cost for a 500-device batch.

% Flagged	Selective (ms)	Encrypt-all (ms)	Savings
5%	53.61	1072.27	95.0%
20%	214.45	1072.27	80.0%
50%	536.13	1072.27	50.0%
100%	1072.27	1072.27	0.0%

entries shown.

A. Failure-Prediction Accuracy

Table 2 reports mean $\pm$ std over 10 independent seeds on a 12,000-window held-out set per seed. Logistic regression reaches a mean AUC of 0.939 (± 0.006) at 0.066 ms (± 0.001) latency; the random forest reaches a comparable 0.929 (± 0.008) AUC but is roughly $45\times$ slower (2.951 ms). Given the edge-latency budget, logistic regression is preferable despite the forest’s marginally higher precision.

B. Selective Encryption Overhead

A single ring-LWE Enc_{QR} operation ($N=256$) costs approximately 2.14 ms over 30 trials – roughly $3.6\times$ cheaper than our retired bit-by-bit plain-LWE prototype, since ring-LWE packs an entire session key per operation instead of one bit. Table 3 reports the cost of encrypting a 500-device batch versus flagged fraction. Because encryption is conditional (Eq. (3)), cost scales with flagged fraction, not fleet size: at the 6–13% flagged fraction observed in our scalability sweep (Section D), selective encryption cuts cost by roughly 87–94% versus encrypting every device – at the stated trade-off that non-flagged telemetry is hash-committed but not confidentiality-protected, by design (Section A).

C. Ablation Study

To isolate how much cost, and which security property, each component contributes, we compare four configurations on the same 200-device batch (25 flagged, 20 trials): baseline predict-only costs 0.28 ms (no confidentiality/tamper-evidence); +selective encryption raises this to 67.92 ms (confidentiality only); +Merkle audit only adds under 1 ms at 0.90 ms (tamper-evidence only); and the full framework costs 68.47 ms (both). Encryption dominates cost by roughly two orders of magnitude over prediction or auditing alone. The two security properties are separable: only encryption-enabled configurations protect confidentiality, and only ledger-enabled ones make commitments tamper-evident.

D. Scalability and On-Chain Storage

Table 4 sweeps batch size from 10 to 800 devices. The on-chain footprint is constant at 32 bytes regardless of batch size,

Table 4: Scalability: edge-cycle time and on-chain footprint vs. fleet size.

Batch	Flagged	Total (ms)	On-chain / raw (bytes)
10	0	0.35	32 / 1,681
100	0	0.46	32 / 17,383
400	26	58.83	32 / 69,532
800	67	148.36	32 / 139,068

Table 5: Hybrid public-key step: ring-LWE vs. classical baselines, same AES-256-GCM payload (30 trials each).

Scheme	Time (ms)	PK-step (B)	PQ assumption
RSA-2048-OAEP	0.049 ± 0.084	256	None (broken by Shor)
ECDH-P256+HKDF	0.126 ± 0.023	33	None (broken by Shor)
Ring-LWE (ours)	2.102 ± 0.033	832	Lattice, demo. params

versus 139,068 bytes of raw payload at 800 devices – a reduction exceeding $4,300\times$. Total edge-cycle time grows from 0.35 ms to 148.36 ms; roughly 98% of that cost is conditional encryption. Comparing edge-local decisions against an edge-plus-cloud baseline paying a modeled round trip (Section IV), edge-local averages 0.150 ms versus 75.29 ms cloud-dependent at our reference 70 ms RTT; sweeping RTT from 10 to 200 ms gives a speedup from roughly $50\times$ to over $1,000\times$. Fig. 2 plots both against batch size.

E. Post-Quantum vs. Classical Baseline

Table 5 compares the ring-LWE hybrid public-key step against the classical primitives it replaces, all wrapping the same AES-256-GCM payload. Ring-LWE is roughly $43\times$ slower than RSA-2048-OAEP and $17\times$ slower than ECDH-P256, with a larger public-key-step output (832 vs. 256/33 bytes). In exchange, RSA-2048 and ECDH-P256 are both broken by Shor’s algorithm, while ring-LWE’s hardness rests on the lattice problems in [36]. Given a single flagged event already costs 2.1 ms end-to-end and encryption is selective, we consider this overhead acceptable for the stated threat model.

F. Cryptographic Functionality and Integrity Validation

We constructed a 50-event batch, committed its Merkle root, and tested three cases: a legitimate inclusion proof (accepted), a proof recomputed against a retroactively forged event (rejected), and a ledger block whose root was altered after the fact (flagged invalid). We separately validated the Phase 3 admission boundary end to end: a validly-signed fresh event was accepted; the same event re-submitted with its captured nonce was rejected (nonce_replay); and a forged/mismatched signature was rejected (invalid_signature) before the replay check ran. Over 100 trials, keypair generation plus signing costs 0.55 ms (± 0.02) and verification 0.15 ms (± 0.01), both negligible next to the 2.1 ms ring-LWE step. A broader suite also passed: ring-LWE round-trip correctness, NTT-vs-schoolbook cross-validation, and Merkle-proof correctness across tree sizes 1–129. These are functional correctness tests, not a security proof (Section VI).

G. Differentiation from State of the Art

Relative to the closest related work (Section II), scoped by what each work describes for itself: edge+PQC work [11] does not condition encryption on the prediction outcome or anchor a ledger; conceptual ML+quantum-computing treatments [12, 18] lack an evaluated implementation; federated blockchain trust frameworks [20] add PQC and a consensus ledger but no prediction-gated encryption, at the cost of a consensus layer we deliberately avoid (Section C). None combines edge-resident prediction, prediction-gated PQC, and Merkle-anchored auditing in one measured pipeline.

VI. SECURITY ANALYSIS

The preceding sections report functional and performance measurements; this section states what they do and do not imply about security.

A. Confidentiality Analysis

Confidentiality covers only telemetry the predictor flags as failure-prone ($I_i(t) = 1$); non-flagged telemetry is hash-committed but sent in the clear (Section B). This makes the gate security-relevant: a *false negative* (a degrading device scored healthy) sets $I_i(t) = 0$, exposing precisely the telemetry most likely to be sensitive – so θ (Eq. (1)) is a security parameter, not only an accuracy one. Sweeping θ on the held-out test split: 0.15 gives 29.3% exposure (1–recall) and a 6.8% encryption rate (138.5 ms/1k events); our deployed point 0.72 ($\alpha=1, \beta=2$) gives 21.9%/7.9% (160.6 ms); and 0.85 gives 18.5%/9.6% (196.0 ms) – exposure and encryption rate move together, so lowering β has a quantified confidentiality cost. Confidentiality of flagged telemetry rests on ring-LWE hardness (Section E) and AES-256-GCM, conditioned on key management our prototype implements but does not harden against local extraction.

B. Integrity Analysis

The Merkle-root/hash-chain design detects *post-commitment* tampering: any change to a committed event or prior block changes a recomputed hash, detected with SHA-256-collision-level probability – exercised by Section F’s forged-event and altered-block tests, both correctly rejected. This says nothing about whether the committed data was truthful when written; Section C addresses that.

C. Compromised Edge Nodes and Zero-Trust Scope

Section A states the design tolerates compromised devices or edge nodes; this needs a precise boundary. A node compromised *before* it computes a batch’s root can fabricate readings and commit an internally-consistent root that passes every check in Section B, since those checks verify consistency, not truthfulness. What the design catches is *subsequent* alteration; our single-writer ledger does not implement the cross-node checks multiple independent writers would enable. The claim is tamper-evidence for honestly-committed records, not detection of a node fabricating self-consistent data at commit time – also why we avoided a consensus ledger (Section G), which raises the cost of forgery but does not eliminate it for an

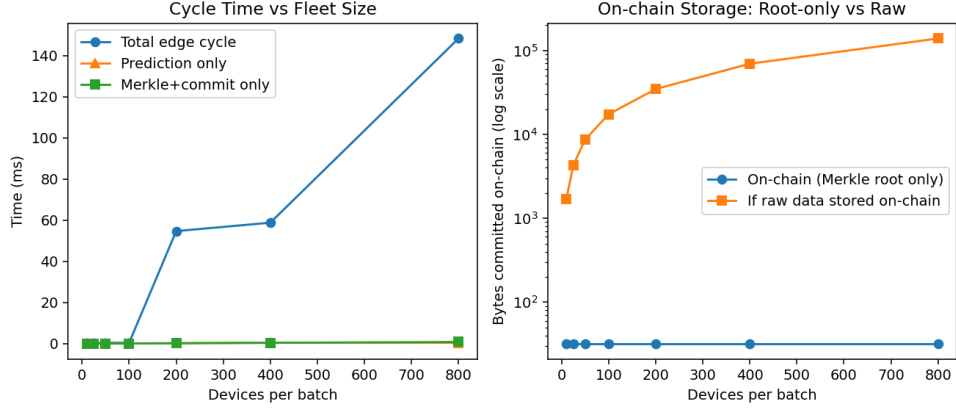


Figure 2: Cycle time (left) and on-chain bytes, log scale (right) vs. batch size.

adversary controlling the sensor.

D. Replay Protection

Two mechanisms enforce replay resistance. The Phase 3 admission check rejects a replayed nonce (`nonce_replay`) or an out-of-window timestamp (`stale_or_future_timestamp`). Independently, since each ledger block’s hash depends on the previous one, a replayed or reordered commitment either duplicates an existing hash or breaks the chain – validated in Section F. The admission check stops replay cheaply before cryptographic work is spent; hash-chaining is an independent second check at the ledger layer.

E. Post-Quantum Security Scope

Four claims are kept separate here. *Assumption*: ring-LWE’s hardness [36] is the same lattice family NIST relies on for ML-KEM/Kyber [7, 16]; we contribute no new hardness result. *Implementation*: our from-scratch Enc_{QR} is cross-validated against an independent schoolbook NTT reference and passes round-trip tests (Section F), but has no formal verification or side-channel hardening. *Security level*: our parameters ($N=256$, $q=7681$) target an estimated 64-bit classical/58-bit quantum level [37], below the 128-bit NIST target; scaling to $N=512/1024$ is a direct, unchanged-algorithm step at higher cost (Section E). *Standardization*: this is the same mathematical family as, but is not, standardized ML-KEM. The contribution is architectural, not a new primitive or standards-track claim.

VII. CONCLUSION AND FUTURE WORK

This paper presented a zero-trust, edge-native framework unifying IoT failure prediction, selective post-quantum encryption, replay-resistant event admission, and blockchain-anchored auditing within a single four-phase pipeline, evaluated on a documented, reproducible implementation. An edge-deployable logistic-regression predictor achieved a mean AUC of 0.939 (± 0.006 over 10 seeds) at sub-millisecond latency; a structured, NTT-validated ring-LWE construction replaced an earlier bit-by-bit prototype at $3.6\times$ lower cost against classical RSA-

2048/ECDH-P256 baselines; conditional encryption cut overhead by roughly 87-94% at observed flagged fractions; and Merkle-root anchoring kept on-chain storage constant regardless of fleet size, with tamper-detection tests confirming forged events and altered blocks are reliably caught. Section VI states what these results do and do not establish: post-commitment tampering is detected with high confidence, but a fully compromised edge node fabricating self-consistent telemetry at commitment time is out of scope for a single-writer ledger; a false negative is a confidentiality exposure, making θ a security parameter; and the post-quantum guarantee is scoped to an assumed-hard lattice problem realized by an uncertified, demonstration-strength implementation.

Limitations: evaluation uses a documented synthetic telemetry generator, since no public dataset matched this work’s multi-device, labeled-onset setting. A partial generalization check – re-evaluating the trained predictor, with no re-fitting, on a second fleet with a higher failure rate, earlier onsets, and noisier sensors – showed AUC dropping only slightly (0.942 to 0.929) while precision fell more (0.946 to 0.878); real fleet telemetry is future work. The ring-LWE construction is from-scratch, pure-Python rather than a certified library, at a demonstration (64-bit/58-bit) rather than Kyber-strength level. The cloud-latency baseline is modeled, and selective encryption leaves non-flagged telemetry confidentiality-unprotected by design. The single-writer ledger leaves consensus throughput, Byzantine fault tolerance, cross-verification, adaptive threshold tuning, and heterogeneous edge hardware for future work.

Conflicts of Interest. The authors declare no conflicts of interest relevant to the content of this article.

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